Solving quadratic equations using sq[redacted] roots

1

a No b Yes c No d Yes

2 $u=\sqrt{d}, u=-\sqrt{d}$

3 $x=-12$

4

a $x=5, x=-5$

b $x=9, x=-9$

c $x=7\sqrt{2}, x=-7\sqrt{2}$

d $x=\sqrt{11}, x=-\sqrt{11}$

e $y=2, y=-2$

f $y=\frac{6}{5}, y=-\frac{6}{5}$

g $x=\sqrt{10}, x=-\sqrt{10}$

h $k=2, k=-2$

i $x=5, x=-5$

j $x=6, x=-6$

k $x=15, x=-15$

l $x=\frac{8}{9}, x=\frac{-8}{9}$

m $x=\frac{6}{5}, x=-\frac{6}{5}$

n $p=12, p=-12$

o $x=16, x=-2$

p $x=4, x=-18$

q $x=7, x=1$

r $x=4+\sqrt{10}, x=4-\sqrt{10}$

s $x=\frac{5}{4}, x=-\frac{11}{4}$

t $x=\sqrt{3}, -\sqrt{3}$

5 None

6 Possibel answers:
$x^2=25$ or $x^2-25=0$

7 $x=\pm\sqrt{\frac{c}{a}}$

8 $x=\pm\frac{6}{c_1}$

9 $x=\pm 4$

10 $t=22, t=14$

Additional questions

11

a $3\sqrt{2}$ **b** $3\sqrt{10}$

12 $t = 9$ seconds

13 $t = 7.4$ seconds

14 $v = 16$ metres/second

15 Eduardo has incorrectly taken the square root of both sides. When taking the square root of something, we need to consider both the positive and the negative case, so an equivalent equation would be $x + 7 = \pm 2\sqrt{7}$. Nicolette has incorrectly subtracted 7 from both sides before taking the square root. The correct equation to solve is $x + 7 = \pm 2\sqrt{7}$ giving an answer of $x = -7 \pm 2\sqrt{(7)}$.

16

a $x = \pm\sqrt{\dfrac{c}{a}}$

b $x = \pm\sqrt{\dfrac{c}{a}}$

c No real solutions.

d $x = 0$